Equivalence of Joint and Sequential Conditioning for Gaussian Processes with Equality Observations and Inequality Constraints

Notation, Definitions, and Proof Roadmap for Proposition 1

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1 Setting and Statement

1.1 The model

Throughout, U is a *separable* Banach (or Hilbert) space of functions carrying a Gaussian prior, and u is a U-valued Gaussian random element. Two finite-dimensional linear observation mechanisms act on u:

- **Equality observations.** A bounded linear operator $L_{\mathrm{eq}} : U \to \mathbb{R}^m$ produces

$$Y \;=\; L_{\mathrm{eq}} u + \varepsilon_{\mathrm{eq}}, \qquad \varepsilon_{\mathrm{eq}} \sim \mathcal{N}(0, \Sigma_{\mathrm{eq}}), \tag{1}$$

  where $\varepsilon_{\mathrm{eq}}$ is independent of u. These model, e.g., PDE collocation residuals, boundary data, or direct point evaluations.

- **Virtual inequality variables.** A bounded linear operator $L_{\mathrm{ineq}} : U \to \mathbb{R}^q$ produces the finite-dimensional virtual variables

$$Z \;=\; L_{\mathrm{ineq}} u + \varepsilon_v, \qquad \varepsilon_v \sim \mathcal{N}(0, \Sigma_v), \tag{2}$$

  with ε_v independent of $(u, \varepsilon_{\mathrm{eq}})$. The inequality information is imposed through the *box event*

$$C \;=\; \{\, a \le Z \le b \,\}, \qquad a, b \in [-\infty, +\infty]^q,\ a < b \text{ componentwise.} \tag{3}$$

Assumption 1. The virtual-noise covariance is strictly positive definite: $\Sigma_v \succ 0$.

Assumption 1 guarantees that $Z \mid Y = y$ is a nondegenerate Gaussian vector, hence $P(C \mid Y = y) > 0$, so all conditional laws appearing below are well defined.

1.2 The sequential construction

The *joint* object of interest is the posterior obtained by conditioning u on both pieces of information at once, $\mathcal{L}(u \mid Y = y,\, C)$. The *sequential* construction instead performs the two updates in order:

(S1) **Equality update.** Fix the observed value $Y = y$ and draw

$$u_{\mathrm{cond}} \sim \mathcal{L}(u \mid Y = y) =: \mu_y.$$

(S2) **Fresh virtual noise.** Independently of u_{cond}, draw

$$\widetilde{\varepsilon}_v \sim \mathcal{N}(0, \Sigma_v).$$

(S3) **Inequality update.** Form the sequential virtual variables and box event

$$Z_{\mathrm{seq}} \;=\; L_{\mathrm{ineq}} u_{\mathrm{cond}} + \widetilde{\varepsilon}_v, \qquad C_{\mathrm{seq}} \;=\; \{\, a \le Z_{\mathrm{seq}} \le b \,\}, \tag{4}$$

and condition u_{cond} on C_{seq}.

1.3 Statement

Proposition 1 (Joint = sequential conditioning)**.** *Under Assumption 1, and with the constructions (1)–(4),*

$$\boxed{\mathcal{L}(u \mid Y = y,\, C) \;=\; \mathcal{L}(u_{\mathrm{cond}} \mid C_{\mathrm{seq}})} \tag{5}$$

as Borel probability measures on U.

Remark 1.1 (What "equality of laws" buys). Identity (5) is an equality of measures on the full Borel σ-algebra $\mathscr{B}(U)$ (Definition 2.2). It is therefore strictly stronger than agreement of posterior means, covariances, or marginals at finitely many prediction points: *every* probabilistic query about the constrained posterior — path probabilities, functionals, exceedance events — gives the same answer under either construction. Computationally, this licenses the standard two-stage scheme: perform the Gaussian equality update in closed form first, then impose the inequality constraints on the resulting posterior (e.g. by sampling the truncated virtual variables).

2 Notation and Measure-Theoretic Preliminaries

This section collects, in one place, every notion used in the proof. Definitions 2.1–2.4 concern probability measures and their transformations; Definitions 2.5–2.7 and Fact 2 supply the machinery for upgrading finite-dimensional agreement to full agreement on $\mathscr{B}(U)$.

2.1 Laws, Borel sets, and equality of measures

Definition 2.1 (Law of a random element)**.** For a random element X taking values in a measurable space $(E, \mathscr{E})$, its *law* (distribution) is the probability measure

$$\mathcal{L}(X)(A) \;=\; P(X \in A), \qquad A \in \mathscr{E}.$$

Conditional laws are denoted analogously; e.g. $\mathcal{L}(u \mid Y = y)$ is the conditional probability measure describing the random function u after observing $Y = y$.

Definition 2.2 (Borel σ-algebra)**.** A *Borel set* of a topological space U is any set obtainable from the open sets by countable unions, countable intersections, and complements. The collection of all Borel subsets of U is the *Borel σ-algebra*, denoted $\mathscr{B}(U)$. A (Borel) probability measure on U assigns probabilities to all sets in $\mathscr{B}(U)$, and two such measures μ_1, μ_2 are *equal* when

$$\mu_1(A) = \mu_2(A) \qquad \text{for every } A \in \mathscr{B}(U).$$

Definition 2.3 (Pushforward measure)**.** Let $T : E \to F$ be measurable and let μ be a probability measure on E. The *pushforward* of μ through T is the measure $T_{\#}\mu$ on F defined by

$$(T_{\#}\mu)(A) \;=\; \mu\big(T^{-1}(A)\big), \qquad A \in \mathscr{B}(F).$$

Equivalently, if $X \sim \mu$ then $T(X) \sim T_{\#}\mu$. The single most-used consequence is:

$$\mathcal{L}(X_1) = \mathcal{L}(X_2) \quad\Longrightarrow\quad \mathcal{L}\big(T(X_1)\big) = \mathcal{L}\big(T(X_2)\big),$$

i.e. *equal laws in, equal pushforward laws out.* In the proof the relevant map is

$$T(v, e) \;=\; \big(L_* v,\; L_{\text{ineq}} v + e\big), \qquad T : U \times \mathbb{R}^q \to \mathbb{R}^p \times \mathbb{R}^q, \tag{6}$$

for a bounded linear $L_* : U \to \mathbb{R}^p$.

Definition 2.4 (Product measure and independence)**.** If $X \sim \mu$ on E and $W \sim \gamma$ on F are *independent*, their joint law on $E \times F$ is the *product measure* $\mu \otimes \gamma$, characterized by

$$(\mu \otimes \gamma)(A \times B) \;=\; \mu(A)\,\gamma(B), \qquad A \in \mathscr{B}(E),\; B \in \mathscr{B}(F).$$

In the proof, the two product measures of interest are both equal to $\mu_y \otimes \gamma_v$, where

$$\mu_y = \mathcal{L}(u \mid Y = y), \qquad \gamma_v = \mathcal{N}(0, \Sigma_v).$$

2.2 Cylinder sets and the Dynkin machinery

Definition 2.5 (Cylinder set)**.** A *cylinder set* in U is a set of the form

$$C_{\ell, B} \;=\; \big\{\, v \in U :\; \big(\ell_1(v), \ldots, \ell_q(v)\big) \in B \,\big\},$$

where $\ell_1, \ldots, \ell_q \in U'$ are bounded linear functionals and $B \in \mathscr{B}(\mathbb{R}^q)$. A cylinder set asks whether *finitely many linear measurements* of v land in a finite-dimensional Borel set.

Example 2.1. If point evaluation is continuous on U (e.g. U an RKHS or a space of continuous functions), then

$$\{\, v :\; v(x_1) \le 1,\; v(x_2) > 0 \,\}$$

is a cylinder set, with $\ell_i(v) = v(x_i)$ and $B = (-\infty, 1] \times (0, \infty)$.

Fact 1 (Cylinder sets generate the Borel σ-algebra)**.** *If U is a separable Banach space, the cylinder sets generate $\mathscr{B}(U)$:*

$$\sigma\big(\{\textit{cylinder sets}\}\big) \;=\; \mathscr{B}(U).$$

Separability is what allows open balls — and hence all open sets — to be recovered from countably many linear functionals.

Definition 2.6 (π-system)**.** A collection $\mathcal{C}$ of subsets is a *π-system* if it is closed under finite intersections:

$$A, B \in \mathcal{C} \quad\Longrightarrow\quad A \cap B \in \mathcal{C}.$$

The cylinder sets form a π-system: intersecting two cylinder sets simply stacks their defining functionals and intersects the corresponding finite-dimensional Borel conditions.

Definition 2.7 (λ-system)**.** A collection $\mathcal{D}$ of subsets of U is a *λ-system* (Dynkin system) if

(i) $U \in \mathcal{D}$;

(ii) $A \in \mathcal{D} \implies A^c \in \mathcal{D}$;

(iii) $\mathcal{D}$ is closed under countable *disjoint* unions.

The prototypical example: for any two probability measures μ_1, μ_2, the agreement class $\{\, A : \mu_1(A) = \mu_2(A) \,\}$ is a λ-system — this follows from $\mu_i(U) = 1$, complementation $\mu_i(A^c) = 1 - \mu_i(A)$, and countable additivity.

Fact 2 (Dynkin's π–λ theorem)**.** *If a λ-system $\mathcal{D}$ contains a π-system $\mathcal{C}$, then it contains the σ-algebra generated by $\mathcal{C}$:*

$$\mathcal{C} \subseteq \mathcal{D} \quad\Longrightarrow\quad \sigma(\mathcal{C}) \subseteq \mathcal{D}.$$

Remark 2.1 (How the machinery is deployed)*.* In the proof of Proposition 1: $\mathcal{C}$ is the collection of cylinder sets (π-system, Definition 2.6); $\mathcal{D}$ is the collection of Borel sets on which the two candidate posteriors agree (λ-system, Definition 2.7); and $\sigma(\mathcal{C}) = \mathscr{B}(U)$ by separability (Fact 1). Hence agreement on all cylinder sets — i.e. on all finite-dimensional linear projections — forces agreement on every Borel set.

3 Proof Roadmap

The proof proceeds in five steps. Steps 1–2 pin down a common starting measure; Step 3 transports it through the map T of (6); Step 4 conditions on the box; Step 5 lifts the resulting finite-dimensional agreement to the full Borel law.

Step 1: Fix a concrete equality-conditioned law

Because the event $\{Y = y\}$ typically has probability zero, one must work with an explicit *regular conditional distribution* rather than a naive ratio of probabilities. Define, in function-space notation, the cross-covariance operator

$$\Phi \;=\; \mathrm{Cov}(u, Y)\; \mathrm{Cov}(Y)^{\dagger},$$

where $\dagger$ denotes the pseudoinverse, and decompose

$$u \;=\; \underbrace{u - \Phi Y}_{=:R} + \Phi Y.$$

The residual $R = u - \Phi Y$ is jointly Gaussian with Y and has vanishing cross-covariance with Y; for jointly Gaussian elements, zero cross-covariance implies independence:

$$R \perp\!\!\!\perp Y.$$

Substituting the observed value y then yields the explicit Gaussian conditional law

$$\boxed{\mu_y \;=\; \mathcal{L}(u \mid Y = y) \;=\; \mathcal{L}\big(u - \Phi Y + \Phi y\big).} \tag{7}$$

This fixes one specific version of the conditional distribution on U for the fixed value y, against which everything downstream is measured.

Step 2: Both constructions start from the same product measure

Since ε_v is independent of (u, Y), conditioning on $Y = y$ leaves its law untouched and preserves independence:

$$\mathcal{L}\big((u, \varepsilon_v) \mid Y = y\big) \;=\; \mu_y \otimes \mathcal{N}(0, \Sigma_v).$$

On the sequential side, $u_{\text{cond}} \sim \mu_y$ and $\widetilde{\varepsilon}_v \sim \mathcal{N}(0, \Sigma_v)$ are independent *by construction*, so

$$\mathcal{L}\big(u_{\text{cond}}, \widetilde{\varepsilon}_v\big) \;=\; \mu_y \otimes \mathcal{N}(0, \Sigma_v).$$

Hence, after the equality update, the joint and sequential constructions are built from *exactly the same* joint probability measure on $U \times \mathbb{R}^q$:

$$\boxed{\mathcal{L}\big((u, \varepsilon_v) \mid Y = y\big) \;=\; \mathcal{L}\big(u_{\text{cond}}, \widetilde{\varepsilon}_v\big) \;=\; \mu_y \otimes \mathcal{N}(0, \Sigma_v).} \tag{8}$$

Step 3: Push both product measures through the same map (transfer identity)

Fix any bounded linear projection $L_* : U \to \mathbb{R}^p$ and recall the measurable map

$$T(v, e) \;=\; \big(L_* v,\; L_{\text{ineq}} v + e\big)$$

from (6). Applying T to the two (identical) product measures of (8) and using Definition 2.3:

$$\mathcal{L}\big(L_* u,\, Z \mid Y = y\big) \;=\; T_{\#}\big(\mu_y \otimes \mathcal{N}(0, \Sigma_v)\big) \;=\; \mathcal{L}\big(L_* u_{\text{cond}},\, Z_{\text{seq}}\big).$$

This is the *transfer identity*:

$$\boxed{\mathcal{L}\big(L_* u,\, Z \mid Y = y\big) \;=\; \mathcal{L}\big(L_* u_{\text{cond}},\, Z_{\text{seq}}\big) \qquad \text{for every bounded linear } L_*.} \tag{9}$$

Note that $Z = L_{\text{ineq}} u + \varepsilon_v$ and $Z_{\text{seq}} = L_{\text{ineq}} u_{\text{cond}} + \widetilde{\varepsilon}_v$ are precisely the second coordinates of T evaluated on the respective pairs.

Step 4: Condition both identical joint laws on the same box

The constraint events depend only on the second coordinate of the pair:

$$C = \{Z \in [a, b]\}, \qquad C_{\text{seq}} = \{Z_{\text{seq}} \in [a, b]\}.$$

By Assumption 1, $Z \mid Y = y$ is nondegenerate Gaussian, so

$$P(Z \in [a, b] \mid Y = y) \;=\; P(Z_{\text{seq}} \in [a, b]) \;>\; 0,$$

(the two probabilities being equal by (9) with trivial L_*), and both conditionings below are well defined. For every Borel set $A \subseteq \mathbb{R}^p$, Bayes' formula and the transfer identity (9) give

$$\begin{aligned} P\big(L_* u \in A \mid Y = y,\, C\big) &= \frac{P\big(L_* u \in A,\ Z \in [a,b] \mid Y = y\big)}{P\big(Z \in [a,b] \mid Y = y\big)} \\ &= \frac{P\big(L_* u_{\mathrm{cond}} \in A,\ Z_{\mathrm{seq}} \in [a,b]\big)}{P\big(Z_{\mathrm{seq}} \in [a,b]\big)} \\ &= P\big(L_* u_{\mathrm{cond}} \in A \mid C_{\mathrm{seq}}\big). \end{aligned}$$

Therefore

$$\boxed{\mathcal{L}\big(L_* u \mid Y = y,\, C\big) \;=\; \mathcal{L}\big(L_* u_{\mathrm{cond}} \mid C_{\mathrm{seq}}\big) \qquad \text{for every bounded linear finite-dimensional } L_*.} \tag{10}$$

In words: after conditioning on the box, *all finite-dimensional linear projections of the two posteriors agree.*

Step 5: Lift finite-dimensional agreement to full measure equality

Abbreviate the two candidate posteriors as

$$\mu_{\mathrm{joint}} \;=\; \mathcal{L}(u \mid Y = y,\, C), \qquad \mu_{\mathrm{seq}} \;=\; \mathcal{L}(u_{\mathrm{cond}} \mid C_{\mathrm{seq}}).$$

Step 4, applied with $L_* = (\ell_1, \ldots, \ell_q)$ ranging over all finite families of bounded linear functionals, says exactly that μ_{joint} and μ_{seq} agree on every cylinder set (Definition 2.5):

$$\mu_{\mathrm{joint}}(C_{\ell,B}) \;=\; \mu_{\mathrm{seq}}(C_{\ell,B}).$$

Define the agreement class

$$\mathcal{D} \;=\; \big\{\, A \in \mathscr{B}(U) :\ \mu_{\mathrm{joint}}(A) = \mu_{\mathrm{seq}}(A) \,\big\}.$$

Then:

(a) $\mathcal{D}$ is a λ-system (Definition 2.7);

(b) the cylinder sets form a π-system contained in $\mathcal{D}$ (Definition 2.6 and Step 4);

(c) since U is separable, the cylinder sets generate $\mathscr{B}(U)$ (Fact 1).

Dynkin's π–λ theorem (Fact 2) then yields $\mathscr{B}(U) \subseteq \mathcal{D}$, i.e.

$$\boxed{\mu_{\mathrm{joint}} \;=\; \mu_{\mathrm{seq}} \quad \text{on } \mathscr{B}(U), \qquad \text{that is,} \qquad \mathcal{L}(u \mid Y = y,\, C) \;=\; \mathcal{L}(u_{\mathrm{cond}} \mid C_{\mathrm{seq}}).} \tag{11}$$

This is precisely the final lifting from finite-dimensional marginals to the full Borel law, completing the proof of Proposition 1. ■

4 Logical Summary

The entire argument compresses into the following chain:

1. Define the equality posterior carefully

$$\mu_y = \mathcal{L}(u \mid Y = y) \;\text{ via }\; u = (u - \Phi Y) + \Phi Y, \quad (u - \Phi Y) \perp\!\!\!\perp Y$$

$$\Downarrow$$

2. Both constructions use the same ingredients

$$(u, \varepsilon_v) \mid Y = y \;\overset{d}{=}\; (u_{\text{cond}}, \widetilde{\varepsilon}_v) \;\sim\; \mu_y \otimes \mathcal{N}(0, \Sigma_v)$$

$$\Downarrow$$

3. Apply the same map $T(v, e) = (L_* v,\; L_{\text{ineq}} v + e)$

same law in $\implies$ same pushforward law out: $\quad (L_* u, Z) \mid Y = y \;\overset{d}{=}\; (L_* u_{\text{cond}}, Z_{\text{seq}})$

$$\Downarrow$$

4. Condition on the same allowed box $[a, b]$

$\Sigma_v \succ 0 \Rightarrow$ box has positive probability $\quad\implies\quad$ all finite-dimensional predictions agree

$$\Downarrow$$

5. Extend to whole functions

cylinder sets generate $\mathscr{B}(U)$ (separability) $+$ Dynkin's π–λ theorem

$$\Downarrow$$

$$\mathcal{L}(u \mid Y = y,\, C) \;=\; \mathcal{L}(u_{\text{cond}} \mid C_{\text{seq}}) \quad \text{on } \mathscr{B}(U)$$

Remark 4.1 (Where each assumption enters). *Separability of* U enters only in Step 5 (Fact 1), to ensure cylinder sets generate $\mathscr{B}(U)$. *Gaussianity* enters in Step 1, where zero cross-covariance upgrades to independence, producing the explicit regular conditional distribution (7). *Independence of the virtual noise* enters in Step 2, making both joint laws the same product measure. $\Sigma_v \succ 0$ (Assumption 1) enters in Step 4, guaranteeing the box event has positive probability so that conditioning is elementary and well posed.